NS-3 Port — Walkthrough

What Was Done

Ported the Network Routing Simulator to NS-3, creating a complete simulation script that lives in its own directory separate from our original C++ simulator.

Files Created

File	Location	Description
routing-sim.cc	`ns-3.48/scratch/routing-sim/`	Full NS-3 simulation (~490 lines)

What It Does

1. **Configurable via command-line** — `--routers`, `--topology`, `--linkDelay`, `--randomDelay`, `--packets`, etc.
2. **4 topology types** — Random, Ring, Mesh, Tree (same algorithms as our C++ simulator)
3. **Real network simulation** — actual UDP/IP packets over point-to-point links
4. **NetAnim visualization** — circle layout, color-coded nodes (green=source, red=dest, blue=others)
5. **FlowMonitor statistics** — per-flow and aggregate stats in table format
6. **MobilityHelper** — ConstantPositionMobilityModel to suppress warnings

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Test Results

All 5 tests passed with **100% packet delivery**:

Test	Routers	Topology	Links	Delay	Delivery	Avg Delay
1	20	Random	39	5ms fixed	100%	10.860 ms
2	10	Ring	10	5ms fixed	100%	5.433 ms
3	15	Tree	14	5ms fixed	100%	16.300 ms
4	8	Mesh	28	5ms fixed	100%	5.400 ms
5	20	Random	39	1-10ms random	100%	12.680 ms

> [!NOTE] Delay values make physical sense:
>
> - **Mesh** has lowest delay (1-hop direct connection)
> - **Ring** is moderate (short path around the ring)

- **Random** is moderate (spanning tree + extra edges = short paths)
- **Tree** has highest delay (must go up and down the binary tree)

How To Use

Run a simulation

```
# Open WSL terminal (Kali Linux)
cd /mnt/c/Users/shrir/Downloads/ns-3.48

# Basic run (20 routers, random topology)
./ns3 run routing-sim

# Custom parameters
./ns3 run "routing-sim --routers=50 --topology=ring --packets=100"

# Random per-link delay
./ns3 run "routing-sim --routers=30 --randomDelay=true --delayMin=1 --
delayMax=10"
```

View in NetAnim

```
~/netanim/build/netanim
# File -> Open -> /mnt/c/Users/shrir/Downloads/ns-3.48/animation.xml
# Press Play to watch packets flow
```

All command-line options

```
--routers=N           Number of routers (default: 20)
--topology=TYPE       random, ring, mesh, tree (default: random)
--dataRate=RATE       Link bandwidth (default: 10Mbps)
--linkDelay=DELAY     Fixed delay per link (default: 5ms)
--packets=N           UDP packets to send (default: 100)
--packetSize=N        Bytes per packet (default: 512)
--source=ID           Source node (default: 0)
--destination=ID      Destination node (-1=last, default: -1)
--randomDelay=BOOL    Random per-link delay (default: false)
--delayMin=MS         Min random delay ms (default: 1.0)
--delayMax=MS         Max random delay ms (default: 10.0)
--seed=N              RNG seed (default: 42)
```